

ELAN LONSTEIN

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TECHNICAL SUMMARY

Hands-on mechanical engineering student with experience in rocketry, robotics fabrication, thermal simulation and avionics integration. Experienced in CAD, CNC machining, prototyping, engineering analysis using COMSOL and ANSYS, and rapid manufacturing and fabrication workflows

Education California State Polytechnic University, Pomona
B.S. Mechanical Engineering
Expected Graduation: May 2028
GPA: 3.47/4.00

Skills

- CAD & Design: SolidWorks, Onshape, Fusion 360
- Simulation & Analysis: COMSOL Multiphysics, ANSYS, OpenRocket
- Manufacturing & Fabrication: CNC machining, manual mill and lathe operation, additive manufacturing, soldering, mechanical assembly, prototype fabrication
- Programming: MATLAB, Python, C++
- Hardware & Electronics: Arduino, Raspberry Pi, sensor integration, avionics systems, accelerometers, altimeters, onboard data acquisition

Projects

ROCKET AVIONICS & FLIGHT SYSTEMS

- Integrated onboard avionics including six-axis accelerometers, altimeters and cameras for rocket flight data acquisition.
- Soldered and assembled electronic subsystems capable of operating under high-vibration launch conditions
- Analyzed flight performance data to evaluate rocket stability and validate simulation predictions
- Designed rocket structures and recovery systems using CAD and OpenRocket
- Integrated recovery systems and validated rocket stability through iterative flight testing and simulation analysis

FRC ROBOTICS MANUFACTURING & DESIGN

- Manufactured precision robot components using CNC machining during competitive robotics build cycles
- Designed and iterated mechanical subsystems under weight, packaging, and manufacturability constraints
- Collaborated in rapid design iterations to improve subsystem reliability and performance
- Produced CAD models and fabrication-ready components for drivetrain and intake assemblies

THERMOELECTRIC SYSTEM MODELING

- Modeled thermoelectric device performance using COMSOL Multiphysics and ANSYS to analyze heat transfer behavior
- Evaluated thermal gradients, material selection, and geometry tradeoffs affecting system efficiency
- Conducted engineering simulations to optimize thermal performance under constrained operating conditions